Project Design Phase-II
Data Flow Diagram & User Stories

Date	28 October 2026
Team ID	En25cs3010377
Project Name	EV Market Performance and Insights
Maximum Marks	4 Marks

Data Flow Diagrams:

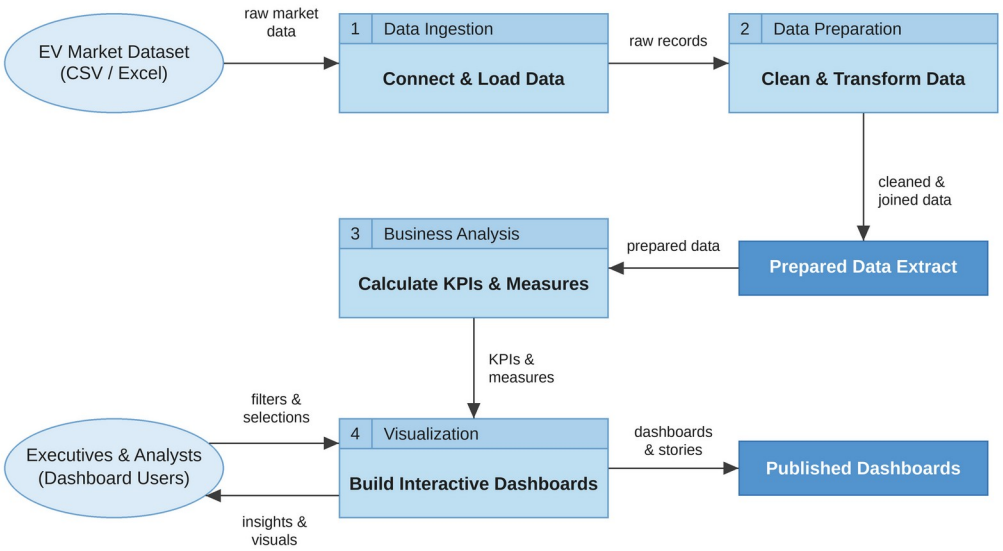
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	**Process**	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	**Data Store**	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	**Data Flow**	The movement of data between entities, processes, and data stores, labeled with what data is

		being passed.

DFD Level 0: EV Market Performance and Insights



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Market Strategy Director	Market Performance Overview	USN-1	As a market strategy director, I can view total EV sales, average vehicle price, average range and CO2 saved as KPI cards, so that I can assess overall market performance at a glance.	KPI cards on Dashboard 1 show correct values and respond to filters	High	Sprint-1
Market Strategy Director	Market Performance Overview	USN-2	As a market strategy director, I can see annual sales units by vehicle category from 2017 to 2027, so that I can identify high-growth segments.	Sales trend chart shows every vehicle category by model year	High	Sprint-1
Market Strategy Director	Customer Satisfaction	USN-3	As a market strategy director, I can compare top manufacturers by customer review and sales units, so that I can judge manufacturer performance and customer satisfaction.	Chart compares manufacturers by customer review and annual sales units	Medium	Sprint-2
Business Intelligence Manager	Manufacturer Analysis	USN-4	As a business intelligence manager, I can compare manufacturers by annual sales and market share, so that I can identify competitive advantages.	Manufacturer matrix ranks manufacturers by annual sales units	High	Sprint-1
Business Intelligence Manager	Production Analysis	USN-5	As a business intelligence manager, I can view production output by country and vehicle category, so that I can understand geographic production trends.	Stacked bar chart shows annual sales units by country, split by vehicle category	High	Sprint-1

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Business Intelligence Manager	Battery Technology Analysis	USN-6	As a business intelligence manager, I can see which battery technologies lead in annual sales, so that I can advise clients on product positioning.	Chart shows annual sales for LFP, NCA, NMC, Sodium-Ion and Solid-State batteries	Medium	Sprint-2
Sustainable Mobility Research Analyst	Battery & Range Analysis	USN-7	As a research analyst, I can compare battery technologies by sales and average battery cycle life, so that I can evaluate technology trends.	Dual-axis chart shows sales bars with a cycle life line for each battery type	High	Sprint-2
Sustainable Mobility Research Analyst	Battery & Range Analysis	USN-8	As a research analyst, I can view the driving range distribution by vehicle category, so that I can assess range improvements.	Box plot shows the range (km) spread for every vehicle category	Medium	Sprint-2
Sustainable Mobility Research Analyst	Charging Performance	USN-9	As a research analyst, I can compare maximum DC charging power with energy efficiency, so that I can study charging performance across categories.	Scatter plot shows charging kW against Wh per km with average reference lines	Medium	Sprint-3
Sustainable Mobility Research Analyst	Environmental Impact	USN-10	As a research analyst, I can see CO2 saved per year by market region and drivetrain, so that I can support sustainability and policy recommendations.	Regional chart shows CO2 saved (tons per year) for AWD, FWD and RWD in every region	High	Sprint-2
Dashboard User (All Roles)	Interaction & Filtering	USN-11	As a dashboard user, I can filter and highlight by vehicle category using the shared legend, so that I can focus on one segment across all charts.	Selecting a category updates every chart on the dashboard	Medium	Sprint-3
Dashboard Administrator	Data Management	USN-12	As a dashboard administrator, I can connect and refresh the EV dataset, so that dashboards always show current data.	Refreshing the data source updates all worksheets without errors	High	Sprint-1